

Regime TVTP with Calibrated Macro with Monotone Heads and VAR(1) Two-Step (v3.1) for Monthly Dynamic Portfolio Construction

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1 Macro overview

We model macro regime evolution with observable regimes and factor-driven sublabels to produce, for each month, full probability vectors for *next month* ($m1$) and *two months ahead* ($m2$) across all composite regimes (macro $\times$ weak flags), plus macro aggregates. The stack is:

- a **time-varying transition probability (TVTP)** macro layer for $P(m_{t+1} \mid m_t, X_t)$, where $m_t \in \{\text{Goldilocks, HeatingUp, SlowGrowth, Stagflation}\}$ is today's macro regime and $X_t = (G_t, I_t, d_t)$ are today's Growth/Inflation trend levels and the current regime duration
- three **monotone heads** for weak Sentiment, weak Liquidity/Credit, and weak Labor, giving $P(\text{flag} \mid m_{\text{dest}}, \text{factors})$, where $m_{\text{dest}} \in \{\text{Goldilocks, HeatingUp, SlowGrowth, Stagflation}\}$ is the destination macro and $\text{factors} = (S_t, \Delta S_t, LQ_t, \Delta LQ_t, LB_t, \Delta LB_t)$ are today's Sentiment/Liquidity/Labor trend levels and 1-month changes (for $m2$ use $t+1$ VAR forecasts in place of t)
- a **VAR(1)**: $Y_{t+1} = a + Y_t\Theta + \varepsilon_{t+1}$ for $Y_t = (G_t, I_t, S_t, LQ_t, LB_t)$, fit on *post-1990* data to avoid pre-1990 structural breaks (data imbalances) and reflect modern dynamics; *ridged* because we ridge-regularize coefficients $\hat{B} = (X^\top X + \lambda R)^{-1} X^\top Y$ (no penalty on the intercept¹) to stabilize estimates and curb overfit, yielding $X_{t+1}^{\text{forecast}}$ for the $m2$ step (with path-wise duration update)

Data & labels. The macro regime $m_t \in \{\text{Goldilocks, Heating Up, Slow Growth, Stagflation}\}$ is determined by the directions of `GrowthIndex_Trend_xt` and `InflationTrend_xt`. The composite label (`EconRegimSpec`) appends *weakSentiment* / *weakLiquidity* / *weakLabor* based on the levels and short-run changes of `SentimentIndex_Trend_xt`, `LiquidityCreditIndex_Trend_xt`, and `LaborIndex_Trend_xt`. Duration is the number of consecutive months spent in m_t .

2 Method (why this stack and how it works)

2.1 TVTP macro layer

We estimate

$$P(m_{t+1} = j \mid m_t = i, X_t) = \frac{\exp\{\beta_{j,0} + \beta_{j,G}G_t + \beta_{j,I}I_t + s_j(d_t)\}}{\sum_k \exp\{\beta_{k,0} + \beta_{k,G}G_t + \beta_{k,I}I_t + s_k(d_t)\}},$$

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¹No penalty on the intercept: shrinking a would distort the unconditional mean and induce level bias; ridge acts only on dynamic coefficients so $E[Y_{t+1}] = a + E[Y_t]\Theta$ remains properly calibrated, especially when regressors are not centered. Regressors are left uncentered because we work in real time on raw index scales. Demeaning would require window/rolling means (risking look-ahead and regime-drift instability) and would shift the intercept, whereas an unpenalized intercept cleanly absorbs the level. Ridge on dynamic coefficients provides stability without altering level calibration.

where G_t and I_t are Growth/Inflation trends, d_t is duration in the current macro, and $s_j(\cdot)$ is a smooth duration effect². We calibrate probabilities for reliability out-of-sample³.

2.2 Monotone factor heads

For each destination macro m , and for each flag $f \in \{\text{weakSentiment}, \text{weakLiquidity}, \text{weakLabor}\}$, we estimate

$$P(f_{t+1} = 1 \mid m_{\text{dest}} = m, \text{factor level}, \Delta\text{factor}),$$

with monotone intent (deteriorating levels $\Rightarrow$ higher weak probability), probability calibration, and empirical-Bayes shrinkage toward macro-conditional base rates to stabilize sparse slices⁴.

2.3 Two-step propagation (VAR + duration paths)

To form m_2 , we must carry information through $t+1$:

1. Forecast the five trends (G, I, S, LQ, LB) with a ridged **VAR(1)** on a modern window (tuned and fixed at window start 1990, ridge $\lambda = 0.05$ ⁵).
2. Update duration path-wise: if $m_{t+1} = m_t$ then $d_{t+1} = d_t + 1$, else $d_{t+1} = 1$.
3. Evaluate $P(m_{t+2} \mid m_{t+1}, X_{t+1}^{\text{forecast}}, d_{t+1})$ and sum over m_{t+1} .
4. Evaluate the heads at $t+2$ using the $t+1$ factor forecasts.

Why this approach. Regime changes are not memoryless and depend on prevailing macro drivers. TVTP captures state- and signal-dependent transitions; heads capture factor-conditioned vulnerabilities; the VAR provides coherent one-step factor forecasts for m_2 . Calibration, shrinkage, and duration deliver stability and interpretability.

3 Turning states into probability vectors (2020–2025)

For month t , the current state is $(m_t, G_t, I_t, S_t, LQ_t, LB_t, \Delta S_t, \Delta LQ_t, \Delta LB_t, d_t)$.

² $s_j(d_t) = \sum_{r=1}^R \gamma_{jr} B_r(d_t)$ is a cubic B-spline of duration with knots at empirical duration quantiles (e.g., 4–6 interior knots) to capture nonlinearity

³After fitting the multinomial TVTP we calibrate class probabilities via classwise *isotonic regression* (a type of regression analysis that finds a non-decreasing (or non-increasing) function that best fits a dataset, minimizing the mean squared error): monotone mapping on a pre-2020 holdout (all observations with Date $\leq$ 2019-12 (1960–2019), reserved for probability calibration (isotonic/Platt) and tuning, while 2020+ is kept strictly for out-of-sample evaluation), CV-smoothed (K-fold, e.g., 5-fold, isotonic calibration using out-of-fold scores, then averaging/refitting the monotone map across folds to reduce variance and avoid overfitting.), with *Platt scaling* $p = \sigma(\text{ascore} + b)$ as fallback when isotonic is unstable or positives are scarce.

⁴For each head and destination macro m , shrink the raw probability $\hat{p}$ toward the macro-conditional base rate $\bar{p}_m$ via a Beta-Binomial prior $a = k\bar{p}_m$, $b = k(1 - \bar{p}_m)$, yielding $\tilde{p} = \frac{a+y}{a+b+n} \approx \frac{k\bar{p}_m + n\hat{p}}{k+n}$ (with n samples, y positives); k (e.g., 30 for Sent/Liq, 60 for Labor) sets shrink strength, stabilizing sparse slices and avoiding overconfident extremes.

⁵Ridge $\lambda = 0.05$ was selected by grid-tuning on a pre-2020 validation slice (2010–2019) to minimize the m_2 log-loss; it provides enough shrinkage to stabilize the VAR(1) without oversmoothing dynamics, with sensitivity showing a shallow optimum around 0.05 (robust for $\lambda \in [0.02, 0.08]$).

3.1 Next month ($m1$): composites and macro

Let a composite label be $c = (m, \mathcal{F})$, where $\mathcal{F}$ is the set of weak flags present in the label. Define the unnormalized mass

$$\tilde{P}_{t+1}(c) = P(m_{t+1} = m \mid m_t, X_t) \times \prod_{f \in \{\text{weakS}, \text{weakLQ}, \text{weakLB}\}} \begin{cases} P(f_{t+1} = 1 \mid m, \text{factors}_t), & f \in \mathcal{F}, \\ 1 - P(f_{t+1} = 1 \mid m, \text{factors}_t), & f \notin \mathcal{F}. \end{cases}$$

Within each macro m , renormalize to match the macro mass exactly⁶:

$$P_{t+1}(c) = \frac{P(m_{t+1} = m \mid m_t, X_t)}{\sum_{c': \text{macro}(c')=m} \tilde{P}_{t+1}(c')} \tilde{P}_{t+1}(c).$$

Macro probabilities are then sums over composites: $P_{t+1}(m) = \sum_{\text{macro}(c)=m} P_{t+1}(c)$.

3.2 Two months ($m2$): composites and macro

First form the macro distribution at $t+2$ by summing over paths m_{t+1} :

$$P_2(m) = \sum_{m_1} P(m_{t+1} = m_1 \mid m_t, X_t) P(m_{t+2} = m \mid m_{t+1} = m_1, X_{t+1}^{\text{VAR}}, d_{t+1}^{\text{path}}).$$

Then compose composites using heads evaluated at $t+2$ (based on the $t+1$ factor forecasts) and renormalize within m exactly as above:

$$\tilde{P}_{t+2}(c) = P_2(m) \times \prod_f [\text{flag prob at } t+2], \quad P_{t+2}(c) = \frac{P_2(m)}{\sum_{c': \text{macro}(c')=m} \tilde{P}_{t+2}(c')} \tilde{P}_{t+2}(c),$$

and $P_{t+2}(m) = \sum_{\text{macro}(c)=m} P_{t+2}(c)$.

3.3 Deliverables (what we export)

For every month t from 2020 onward we produce two wide matrices:

- $m1$: a row vector $P_{t+1}(c)$ over all composite labels c (sums to 1), plus macro aggregation $P_{t+1}(m)$.
- $m2$: a row vector $P_{t+2}(c)$ over all composite labels (sums to 1), plus macro aggregation $P_{t+2}(m)$.

These are saved as two Excel files (one for $m1$, one for $m2$), one row per month and one column per composite regime.

4 Performance (2020+ diagnostics)

- Macro-only TVTP: out-of-sample (OOS) log-loss ≈ 0.63 ; Top-1 accuracy ≈ 0.86 ⁷.

⁶The within-macro renormalization enforces mass conservation so that composites under macro m sum exactly to $P(m_{t+1} = m)$ from the TVTP; it also corrects for (i) missing/omitted composite labels and (ii) small calibration errors in the heads that would otherwise leak or deficit probability within m .

⁷Macro-only TVTP performance (2020+ OOS): log-loss ≈ 0.63 (lower is better: compared with the uniform baseline $\ln 4 \approx 1.386$) and Top-1 accuracy ≈ 0.86 (the highest-probability macro matches the realized macro in 86% of months).

- Composite $m1$: out-of-sample log-loss ≈ 1.50 (well below $\ln 18 \approx 2.89$ uniform baseline on frequent labels).
- Composite $m2$: out-of-sample log-loss ≈ 1.96 with VAR(1) tuned at window start 1990 and ridge 0.05.
- Sanity: head probabilities are calibrated and shrunk toward macro-conditional base rates; within each macro, composite probabilities renormalize to the macro mass exactly.

5 Robustness and safeguards

- **Structural change.** VAR fitted on a recent window with ridge shrinks instability; TVTP includes duration, allowing learned persistence/mean reversion.
- **Rare labels.** Evaluation focuses on composites with ≥ 12 occurrences to avoid noise; full distributions are still produced; empirical-Bayes shrinkage stabilizes sparse macro slices.
- **Missing data.** If a factor is missing at t , heads revert to macro-conditional base rates; the VAR imputes with training means for one-step forecasts only.
- **Interpretability.** Macro coefficients show how Growth/Inflation tilt transitions; heads reflect monotone vulnerability; any composite probability decomposes into macro step $\times$ flag step.
- **Monitoring.** Reliability curves/Brier scores, rolling OOS log-loss, and drift alerts (e.g., head probabilities > 0.9 or < 0.1 for extended periods) keep the system in check.

6 How to use the vectors

- **Narrative.** “Given today’s state, next-month macro probabilities are ...; two months out are ...; conditional weaknesses under each macro are ...”
- **Portfolio mapping.** Tilt exposure where $m2$ mass concentrates (turn the hogest-probability regimes into weights to overweight specific assets); overlay weak-flag odds (e.g., weakLiquidity) for risk-sensitive allocation.
- **Attribution.** Decompose spikes in a composite probability into macro persistence/rotation vs. factor-driven weakness via heads. Decompose a change in a composite $c = (m, \mathcal{F})$ into macro vs. head effects:

$$\Delta P(c) \approx \underbrace{\Delta P(m) H_t}_{\text{macro persistence/rotation via TVTP}} + \underbrace{P_t(m) \Delta H}_{\text{factor weakness via heads}}.$$

$$H_t = \prod_{f \in \mathcal{F}} q_t^{(m)}(f) \prod_{f \notin \mathcal{F}} [1 - q_t^{(m)}(f)], \quad q_t^{(m)}(f) = \Pr(f_{t+1} = 1 \mid m, \text{factors}_t).$$

- **Sensitivity.** Compare baseline vs. counterfactual duration (reset $d_t = 1$) to quantify how time-in-state shifts path probabilities.

7 Dynamic Monthly Portfolio Construction (TVTP Regimes + Trend Filters)

We build a dynamic, monthly-rebalanced portfolio driven by time-varying transition probabilities (TVTP) over economic regimes. For each as-of month t , we read the regime probability vector $p_r(t)$ from the M1 file. For every regime r , we estimate each ticker’s *adjusted return* as a Sharpe ratio using only history strictly prior to t and only rows where the resolved regime equals r : monthly *log* returns are annualized (mean/std). Tickers with fewer than MIN_OBS_PER_REGIME months in that regime are discarded; with POSITIVE_ONLY we also drop non-positive Sharpe.

Eligibility is then gated by trend filters measured at t . For indices/sectors/commodities we use the trend columns that literally contain “trend” in the Market and FI trend files; the rule is *positive and improving* (current > 0 and current > prior by TREND_DELTA_LAG). For *stocks* we also require a price-trend test based on the exact `ticker_P` column in the Market trend file (this column name does not include the word “trend”); the default price rule is *improving or positive*. The trading universe is fixed:

Equities: S5HLTH_P, S5ENRS_P, S5RLST_P, S5UTIL_P, S5CONS_P, SPW_P, RLV_P, SPX_P, S5INFT_P, S5TELS_P, S5COND_P, S5FINL_P, S5INDU_P. *Commodities*: CO1, HG1, XAUUSD. *Fixed Income*: UST1-3, UST20PLUS, UST3-5, UST5-7, UST3-10, UST5-10, UST3-7, USTALL, USAGG1-3, USAGG5-10, IG1-5, IG5-10, IG10PLUS, USHY, USHY1-5, HY0-5, MBS.

If a regime r has no eligible tickers (e.g., sparse history or failed trends), we *do not* borrow across different roots. Instead, we walk *same-root fallbacks* by dropping modifiers stepwise (e.g., `root_a_b_c` → `root_a_b` → `root_a` → `root`) and use the closest fallback with eligibles. If even the root has none, that regime’s probability is left *uninvested* (unless REALLOCATE_MISSING is enabled, which rescales invested sleeves to sum to one). We record the fraction actually deployed as *InvestedProb*.

Weights are formed *within* each (chosen) regime: its probability $p_r(t)$ is split across that regime’s eligible tickers in proportion to Sharpe^{POWER} (with POWER=1 by default; higher concentrates more in the best names). The portfolio for month $t+1$ uses weights computed at t . Realized portfolio return in $t+1$ is the weighted sum of *arithmetic* returns; we chain monthly P&L to produce Equity, CAGR, volatility, Sharpe, and max drawdown.

For analysis without rerunning the backtest, we save a *weights CSV* (as-of t) and then build a *monthly analytics* dataset by joining those weights to next-month returns and tagging each ticker to an asset class (Equities/Commodities/Fixed Income). From this we derive per-month class sleeves (weight, contribution, sleeve return), the total portfolio return, and *InvestedProb*; yearly aggregations power the interactive views: (i) yearly returns by asset class, (ii) yearly Sharpe by asset class, (iii) stacked average yearly weights (normalized to 100%) with class returns on a secondary axis, and (iv) per-year tables of the highest-return and longest-held tickers by asset class (computed over months actually held).